

Linear Applications and Matrices

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Linear applications

Definition

Let E and E' be two Vector Spaces over $\mathbb{K}$ and f an application from E to E' ($f : E \rightarrow E'$). Then, we say that f is **linear** if

- ① $\forall u, v \in E, f(u + v) = f(u) + f(v)$;
- ② $\forall u \in E, \forall \lambda \in \mathbb{K}, f(\lambda \cdot u) = \lambda \cdot f(u)$.

Remarks :

- If f is linear, then $f(0_E) = 0_{E'}$;
- The set of linear applications from E to E' is denoted $\mathcal{L}(E, E')$;
- A linear application is an application that respects the operations $(+, \cdot)$ of a Vector Space.
→ A linear application allows the passage from a Vector Space to an other one.

Examples :

- For every $\mathbb{K}$ -Vector Spaces E, F , the application $f : E \rightarrow F$ defined as $\forall u \in E, f(u) = 0_F$ is linear ;
- For every $\mathbb{K}$ -Vector Spaces E , the application **identity** $Id : E \rightarrow E$ defined as $\forall u \in E, Id(u) = u$ is linear.
- The application **differentiation** $f : \mathbb{K}[X] \rightarrow \mathbb{K}[X]$ defined as $\forall P \in \mathbb{K}[X], f(P) = P'$ is linear.

Exercise :

Determine which of the following applications is linear :

1) $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \forall u = (x_1, x_2, x_3) \in \mathbb{R}^3, f(u) = (2x_1 + x_2, x_2 - x_3),$

2) $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \forall u = (x_1, x_2, x_3) \in \mathbb{R}^3, f(u) = (2x_1x_2, x_2 + 3x_1),$

3) $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \forall u = (x_1, x_2, x_3) \in \mathbb{R}^3, f(u) = (2x_1 + x_2 + x_3 + 5, x_2 - 3x_1),$

4) $f : \mathbb{R}^3 \rightarrow \mathbb{R}, \forall u = (x_1, x_2, x_3) \in \mathbb{R}^3, f(u) = -x_1 + 2x_2 + 5x_3.$

Linear applications in $\mathbb{R}^n$

Definition

- Let

$$f_1 : \mathbb{R}^p \rightarrow \mathbb{R}, f_2 : \mathbb{R}^p \rightarrow \mathbb{R}, \dots, f_n : \mathbb{R}^p \rightarrow \mathbb{R},$$

n functions of p variables. Every function f_i is defined as follows

$$f_i : (x_1, \dots, x_p) \rightarrow f_i(x_1, \dots, x_p) \in \mathbb{R}.$$

- We construct the application $f : \mathbb{R}^p \rightarrow \mathbb{R}^n$ defined as

$$f(x_1, \dots, x_p) = (f_1(x_1, \dots, x_p), \dots, f_n(x_1, \dots, x_p)).$$

- An application $f : \mathbb{R}^p \rightarrow \mathbb{R}^n$ defined as $f(x_1, \dots, x_p) = (y_1, \dots, y_n)$ is said to be **linear** if

$$f \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} a_{11} & \dots & a_{1p} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{np} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix}$$

- By letting $X = (x_1, \dots, x_p)^T$ and $A = (a_{ij}) \in M_{np}(\mathbb{R})$, we obtain

$$f(X) = A \times X$$

Where the matrix A is called the **matrix of the linear application**.

Examples of linear applications in $\mathbb{R}^n$:

1) Reflection with respect to the y-axis :

The linear application

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -x \\ y \end{pmatrix}$$

is the reflection with respect to the y-axis, and its matrix is $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

2) Reflection with respect to the x-axis :

The linear application

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x \\ -y \end{pmatrix}$$

is the reflection with respect to the x-axis, and its matrix is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

3) Reflection with respect to the line $y = x$:

The linear application

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} y \\ x \end{pmatrix}$$

is the reflection with respect to the line $y = x$, and its matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

4) Homothety :

The linear application

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} \lambda x \\ \lambda y \end{pmatrix}$$

is the homothety of the vector (x, y) , and its matrix is $\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$.

5) Rotation :

The linear application

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos(\alpha) \\ r \sin(\alpha) \end{pmatrix} \rightarrow \begin{pmatrix} r \cos(\alpha + \theta) \\ r \sin(\alpha + \theta) \end{pmatrix}$$

is the rotation of angle θ , and its matrix is $\begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$.

6) Orthogonal projection :

- The linear application

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x \\ 0 \end{pmatrix}$$

is the orthogonal projection over the x-axis, and its matrix is $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

- The linear application

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

is the orthogonal projection over the (xy)-plane, and its matrix is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

- Similarly, the orthogonal projection over the (xz)-plane, resp. the (yz)-plane, is of matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, resp. $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

7) Reflection in the space :

- The linear application

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ -z \end{pmatrix}$$

is the reflection with respect to (xy)-plane, and its matrix is $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

- Similarly, the reflection with respect to the (xz)-plane, resp. the (yz)-plane, is of matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, resp. $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Properties of Linear Applications

Endomorphism

Definition

Let E be a $\mathbb{K}$ -Vector Space, we call **endomorphism** of E every linear application from E to E (same starting and ending V.S.). The set of all endomorphisms of E is denoted $\text{End}(E)$ or $\mathcal{L}(E)$.

Examples :

- The linear application identity $\forall u \in E, Id_E(u) = u$ is an endomorphism.
- The linear application $\forall u = (x, y) \in \mathbb{R}^2, f(u) = (y, x)$ is an endomorphism.

Image and Kernel of a linear application

Proposition

- Let E, E' be two Vector Spaces, f a linear application from E to E' and F a Vector Subspace of E . Then $f(F)$ is a Vector Subspace of E' .
- In particular, $f(E)$ is a Vector Subspace of E' called the **image** of f and denoted $\text{Im}(f)$,

$$\text{Im}(f) = f(E)$$

Definition

We call the **rank** of f , denoted $\text{rank}(f)$, the dimension of the image of f ,

$$\text{rank}(f) = \dim(\text{Im}(f))$$

Proposition

Let $f \in \mathcal{L}(E, E')$, then the set defined as

$$\text{Ker}(f) = \{x \in E \mid f(x) = 0_{E'}\}$$

is a Vector Subspace of E called the **kernel** of f .

Proposition 1

Let E be a Vector Space of finite dimension. Then,

- $\text{Im}(f) = f(E)$ is a Vector Space of finite dimension,
- If $(e_1, \dots, e_n)$ is a basis of E , then $\text{Im}(f) = \text{Vect}(f(e_1), \dots, f(e_n))$.

Proposition 2

Let E, E' be two $\mathbb{K}$ -Vector Spaces of finite dimension and $f : E \rightarrow E'$ a linear application. Then, we have

$$\text{rank}(f) \leq \min(\dim E, \dim E')$$

Exercise :

1) Determine the image and kernel of the following function f :

a) f is the function defined from $\mathbb{R}^3$ to $\mathbb{R}^3$ as :

$$f(x, y, z) = (x', y', z'), \text{ where } \begin{cases} x' = x + y - z \\ y' = 2x + y - 3z \\ z' = 3x + 2y - 4z \end{cases}$$

b) f is the function differentiation defined from $\mathbb{R}_n[X]$ to $\mathbb{R}_n[X]$, with $n \in \mathbb{N}^*$, as $f(P) = P'$.

2) Let f be the function defined from $\mathbb{R}^4$ to $\mathbb{R}^3$ as :

$$f(x, y, z, t) = (x', y', z'), \text{ where } \begin{cases} x' = x - y + z \\ y' = 2x + 2y + 6z + 4t \\ z' = -x - 2z - t \end{cases}$$

Determine the rank of f and the dimension of the kernel of f .

Proposition 1

Let $f \in \mathcal{L}(E, E')$. Then, f is injective if and only if $\text{Ker}(f) = \{0_E\}$.

Proposition 2

Let $f \in \mathcal{L}(E, E')$ and $S = \{v_i\}_{i=1\dots p}$ a set of p vectors in E . Then

- ① if f is injective and S is linearly independent, then the set $\{f(v_i)\}_{i=1\dots p}$ is linearly independent in E' .
- ② if f is surjective and S is a generating set of E , then the set $\{f(v_i)\}_{i=1\dots p}$ is a generating set of E' .
- ③ if f is bijective then the image of a basis of E by f is a basis of E' .

Isomorphism

Definition

- A application is said to be an **isomorphism** if it is linear and bijective.
- If $f : E \rightarrow E'$ is an isomorphism, then $f^{-1} : E' \rightarrow E$ is also an isomorphism.

Proposition

Let $f : E \rightarrow E'$ an isomorphism of Vector Spaces. Then, if E (respectively E') is of finite dimension, it implies that E' (respectively E) is also of finite dimension and $\dim(E) = \dim(E')$.

Rank Theorem

Let E, E' be two Vector Spaces of finite dimension and $f : E \rightarrow E'$ a linear application. If $\dim(E) = \dim(E')$, then

$$\dim(E) = \dim(\text{Im}(f)) + \dim(\text{Ker}(f)) = \text{rank}(f) + \dim(\text{Ker}(f))$$

Theorem

Let $f \in \mathcal{L}(E, E')$ a linear application and E, E' are of finite dimension.

If $\dim(E) = \dim(E')$, then the following statements :

- ① f is bijective,
- ② f is injective,
- ③ f is surjective,

are equivalent. I.e.,

$$(1) \Leftrightarrow (2) \Leftrightarrow (3)$$

Exercise :

1) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x, y) = (x - y, x + y)$. Show that f is bijective.

2) Let f be the function defined from $\mathbb{R}_3[x]$ to $\mathbb{R}^4$ as :

$$\forall P \in \mathbb{R}_3[x], f(P) = (P(0), P(1), P(2), P(3))$$

Show that f is an isomorphism.

END

References :

- Joseph Grifphone, *ALGÈBRE LINÉAIRE*, 5e edition.
- Arnaud Bodin et al., *Exo7*, Université de Lille.